Punycode.js

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Punycode.js is a robust Punycode converter that fully complies to RFC 3492 and RFC 5891.

This JavaScript library is the result of comparing, optimizing and documenting different open-source implementations of the Punycode algorithm:

- The C example code from RFC 3492
- `punycode.c` by *Markus W. Scherer* (IBM)
- `punycode.c` by *Ben Noordhuis*
- JavaScript implementation by *some*

- `punycode.js` by *Ben Noordhuis* (note: not fully compliant)

This project was bundled with Node.js from v0.6.2+ until v7 (soft-deprecated).

This project provides a CommonJS module that uses ES2015+ features and JavaScript module, which work in modern Node.js versions and browsers. For the old Punycode.js version that offers the same functionality in a UMD build with support for older pre-ES2015 runtimes, including Rhino, Ringo, and Narwhal, see v1.4.1.

Installation

Via npm:

```
npm install punycode --save
```

In Node.js:

> ⚠️ Note that userland modules don't hide core modules. For example, `require('punycode')` still imports the deprecated core

> module even if you executed `npm install punycode`. Use `require('punycode/')` to import userland modules rather than core modules.

```
const punycode = require('punycode/');
```


API

`punycode.decode(string)`

Converts a Punycode string of ASCII symbols to a string of Unicode symbols.

```
// decode domain name parts
punycode.decode('maana-pta'); // 'mañana'
punycode.decode('--dqo34k'); // '☃-⌘'
```


`punycode.encode(string)`

Converts a string of Unicode symbols to a Punycode string of ASCII symbols.

```
// encode domain name parts
punycode.encode('mañana'); // 'maana-pta'
punycode.encode('☃-⌘'); // '--dqo34k'
```


`punycode.toUnicode(input)`

Converts a Punycode string representing a domain name or an email address to Unicode. Only the Punycoded parts of the input will be converted, i.e. it doesn't matter if you call it on a string that has already been converted to Unicode.

```
// decode domain names
punycode.toUnicode('xn--maana-pta.com');
// → 'mañana.com'
```



```
punycode.toUnicode('xn----dqo34k.com');
// → '☃-⌘.com'

// decode email addresses
punycode.toUnicode('джумла@xn--p-8sbkgc5ag7bhce.xn--ba-lmcq');
// → 'джумла@джpумлатест.bрфa'
```


`punycode.toASCII(input)`

Converts a lowercased Unicode string representing a domain name or an email address to Punycode. Only the non-ASCII parts of the input will be converted, i.e. it doesn't matter if you call it with a domain that's already in ASCII.

```
// encode domain names
punycode.toASCII('mañana.com');
// → 'xn--maana-pta.com'
punycode.toASCII('☃-⌘.com');
```



```
// → 'xn----dqo34k.com'

// encode email addresses
punycode.toASCII('джумла@джpумлатест.bрфa');
// → 'джумла@xn--p-8sbkgc5ag7bhce.xn--ba-lmcq'
```


`punycode.ucs2`

`punycode.ucs2.decode(string)`

Creates an array containing the numeric code point values of each Unicode symbol in the string. While JavaScript uses UCS-2 internally, this function will convert a pair of surrogate halves (each of which UCS-2 exposes as separate characters) into a single code point, matching UTF-16.


```
punycode.ucs2.decode('abc');
// → [0x61, 0x62, 0x63]
// surrogate pair for U+1D306 TETRAGRAM FOR CENTRE:
punycode.ucs2.decode('\uD834\uDF06');
// → [0x1D306]
```

`punycode.ucs2.encode(codePoints)`

Creates a string based on an array of numeric code point values.


```
punycode.ucs2.encode([0x61, 0x62, 0x63]);
// → 'abc'
punycode.ucs2.encode([0x1D306]);
// → '\uD834\uDF06'
```

`punycode.version`

A string representing the current Punycode.js version number.

For maintainers

How to publish a new release

1. On the `main` branch, bump the version number in `package.json`:

   ```
   npm version patch -m 'Release v%s'
   ```

   Instead of `patch`, use `minor` or `major` as needed.

   Note that this produces a Git commit + tag.
2. Push the release commit and tag:

```
git push && git push --tags
```

Our CI then automatically publishes the new release to npm, under both the `punycode` and `punycode.js` names.

Author

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